

Abirami Sridharan

Robotics & Embedded Systems Engineer | ADAS Validation | Autonomous Systems Integration



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Professional Summary

Embedded Systems and DevOps Engineer with experience in embedded software validation, CI/CD automation, robotics integration, and multi-controller embedded platforms. Skilled in Python, C++, Bash, Docker, Azure DevOps, HIL testing, and automation infrastructure for embedded and microcontroller-based systems. Experienced in embedded Linux environments, distributed debugging, telemetry/logging workflows, and system-level validation across automotive ADAS and robotics domains.

Skills

Programming

C, C++, Python, javascript, Bash

Development Tools

Docker, Jenkins, Azure DevOps, Git, GitLab, Lauterbach Trace32

Embedded Systems

Embedded Linux, Automotive Embedded Software, Microcontrollers (TI Controllers, RL78), Real-time Systems, Sensor Integration

Automation & Testing

Test Automation Frameworks, CI/CD Pipelines, Hardware-in-the-Loop Testing

Communication Protocols

SOME/IP, TCP/IP, UDP, I2C, SPI, LIN, Modbus

Robotics & ADAS

ADAS Camera Systems, Radar Validation, Robotics System Integration, Robot Accuracy Calibration, ROS2

Professional Experience

12/2024 – Present
Munich

RobCo GmbH, Robotics Systems Engineer

- Developed HIL-based validation workflows for robotic automation platforms using Linux-based embedded environments.
- Integrated robotic arm systems, control units, and Modbus TCP communication interfaces for system-level validation and diagnostics.
- Developed Python-based automation and telemetry tools for robotic diagnostics, logging, data analysis, and reporting workflows.
- Performed robotic TCP calibration, coordinate transformation analysis, and system accuracy validation for robotic positioning workflows.

06/2022 – 11/2024
Leonberg

Robert Bosch GmbH, Test Automation & DevOps Engineer

- Developed automated validation frameworks for unit, integration, and code coverage testing across Radar Gen6/Gen7 embedded controller platforms supporting ISO 26262-aligned verification workflows.
- Implemented CI/CD pipelines in Azure DevOps for automated validation across multiple hardware and embedded controller variants.
- Developed automated validation workflows for distributed multi-controller embedded systems, improving validation coverage, release reliability, and system integration efficiency.

	• Debugged embedded software and low-level system issues using Lauterbach Trace32 and distributed debugging workflows.
05/2020 – 05/2021 Reutlingen	Technology & Strategy GmbH(Consultant for Bosch), <i>Test Automation & DevOps Engineer</i> • Developed a C++14 validation framework for battery and drive unit embedded systems. • Designed automated CI/CD validation pipelines using Ansible for continuous test execution. • Implemented unit and integration testing using Google Test to improve software quality and validation coverage.
08/2018 – 01/2020 Abstatt	Technology & Strategy GmbH (Consultant for Bosch), <i>ADAS Camera System Developer</i> • Developed software features for multi-range ADAS camera perception systems. • Implemented SOME/IP communication between Zynq and i.MX6 processors for distributed embedded perception platforms. • Developed ADTF filters for perception data visualization, debugging, and analysis workflows. • Contributed to perception pipeline integration, embedded debugging, and system-level validation of ADAS camera platforms.
07/2017 – 07/2018 Bremen	HELLA GmbH, <i>Master's Thesis Embedded Software for Automotive CO₂ Sensor Systems</i> • Developed embedded software for RL78 microcontroller-based automotive CO₂ sensor systems. • Implemented PID-based temperature control algorithms for stable sensor operation. • Worked with embedded communication protocols including I2C, SPI, and LIN.
10/2016 – 06/2017 Hannover	WABCO GmbH, ADAS Test Framework Intern • Developed automation features for ADAS validation and test frameworks. • Worked with calibration data using MATLAB/Simulink and TargetLink. • Validated ADAS control functions including Adaptive Cruise Control (ACC) and ABS.

Projects

Edge IoT Humidity Monitoring Platform [↗](#)

Developed a Raspberry Pi-based IoT monitoring platform using Python, Docker, FastAPI, BLE communication, and automated reporting workflows.

Sign Language Detection [↗](#)

Developed a real-time computer vision prototype using MediaPipe and machine learning for gesture recognition and edge-device inference workflows.

Smart Parking Detection System (ROS2) [↗](#)

ROS2-based smart parking occupancy detection prototype using computer vision and edge processing.

Education

10/2015 – 08/2018 Chemnitz	M.Sc. Automotive Software Engineering, Technische Universität Chemnitz
05/2009 – 06/2013 India	B.E. Electrical & Electronics Engineering, SCSVMV university